

OSINT RECONNAISSANCE AUTOMATION

Security Intelligence & Attack Surface Analysis Report

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STATUS: COMPLETE / VALIDATED

1. Executive Summary

This project involved the development and deployment of **OSINT_final.py**, an automated reconnaissance tool designed to map the external attack surface of a target domain. By centralizing data from WHOIS registries and the Shodan global IoT scanner, the tool identifies critical exposure points, such as unencrypted services and unauthorized open ports, providing a comprehensive "Attacker's View" of the infrastructure.

2. Technical Methodology

The reconnaissance engine utilizes a multi-threaded logic to gather intelligence across two primary vectors:

- Domain Metadata:** Fetches `python-whois` records to identify administrative contacts, name server configurations, and domain expiration dates.
- Infrastructure Scanning:** Integrates the **Shodan API** to resolve hostnames to IP addresses and extract detailed host reports including open ports, banners, and geolocation.

3. Security & Stealth Mechanisms

To bypass basic Web Application Firewalls (WAFs) and automated bot detection, the script implements:

- **User-Agent Spoofing:** Mimics legitimate Chrome/Windows traffic.
- **JSON Data Persistence:** Results are automatically formatted into `osint_report.json` for immediate ingestion into SIEM or Incident Response platforms.

4. Strategic Recommendations

Based on the reconnaissance capabilities demonstrated, the organization should:

1. Implement weekly automated scans to detect **Shadow IT**.
2. Enforce strict firewall rules for non-standard ports identified by the Shodan engine.
3. Audit WHOIS records to ensure privacy protections are enabled for sensitive administrative data.